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Omnivest

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Resumo

Sarhan Giberti, Luca; Garcia, Igor Scapin Francisco. **Omnivest**. 2025. nº de folhas. TCC (Graduação) – Curso Engenharia de Hardware, Instituto de Tecnologia e Liderança, São Paulo, 2025.

Este trabalho apresenta o desenvolvimento do Omnivest, uma plataforma computacional orientada ao apoio à tomada de decisão de investidores pessoa física no mercado de capitais brasileiro, por meio da análise automatizada de dados financeiros e da aplicação de técnicas de inteligência artificial. O objeto de estudo consiste na concepção, implementação e validação de uma solução tecnológica capaz de extrair, estruturar e analisar informações provenientes de demonstrações financeiras corporativas, bem como de gerar previsões de preços de ações com base em fundamentos econômicos. O principal objetivo do projeto é desenvolver um sistema integrado que reduza a complexidade da análise fundamentalista e amplie o acesso a ferramentas quantitativas normalmente restritas a ambientes profissionais. Para atingir esse objetivo, adotou-se uma abordagem metodológica baseada em desenvolvimento incremental, estruturada em módulos e sprints, combinando práticas de engenharia de software, engenharia de dados e aprendizado de máquina. A solução foi implementada como uma aplicação web full stack, com frontend desenvolvido em React e backend baseado em FastAPI, incluindo autenticação segura, visualização de métricas financeiras e um serviço de previsão de preços treinado a partir de dados históricos extraídos de relatórios corporativos. O pipeline de dados contempla etapas de coleta, limpeza, padronização e armazenamento de indicadores financeiros, permitindo análises comparativas entre empresas e períodos distintos. Como resultados, obteve-se um protótipo funcional que integra visualizações interativas, indicadores financeiros estruturados e previsões de preços para diferentes horizontes temporais, demonstrando a viabilidade técnica da proposta. Além disso, o projeto desenvolveu um plano de negócios inicial, contemplando análise de mercado, definição do modelo de receita, estratégias de aquisição de usuários e projeções financeiras, evidenciando o potencial de viabilidade econômica da solução. Conclui-se que o Omnivest atende aos objetivos propostos, validando a aplicação de técnicas de inteligência artificial e engenharia de software no contexto de apoio à decisão financeira, e estabelece bases sólidas para futuras evoluções e expansão da plataforma.

Palavras-Chave: inteligência artificial; análise fundamentalista; mercado financeiro; engenharia da computação; sistemas de apoio à decisão.

ABSTRACT

Sarhan Giberti , Luca; Garcia, Igor Scapin Francisco. **Omnivest**. 2025. 16 pages. Final course project (Bachelor) – Course Hardware Engineering, Institute of Technology and Leadership , São Paulo, 2025.

This work presents the development of Omnivest, a computational platform designed to support decision-making for retail investors in the Brazilian capital market through the automated analysis of financial data and the application of artificial intelligence techniques. The object of study consists of the conception, implementation, and validation of a technological solution capable of extracting, structuring, and analyzing information from corporate financial statements, as well as generating stock price forecasts based on economic fundamentals. The main objective of the project is to develop an integrated system that reduces the complexity of fundamental analysis and broadens access to quantitative tools typically restricted to professional investment environments. To achieve this objective, an incremental and iterative development methodology was adopted, structured in modules and sprints and combining principles of computer engineering, software development, data engineering, and machine learning. The solution was implemented as a full-stack web application, with a frontend developed using React and a backend based on FastAPI, incorporating secure authentication mechanisms, interactive visualization of financial indicators, and a prediction service trained on historical data extracted from official corporate reports. The data pipeline includes stages of data collection, cleaning, normalization, and storage, enabling comparative analysis across different companies and time periods. As results, a functional prototype was obtained that integrates structured financial indicators, interactive dashboards, and multi-horizon stock price forecasts, demonstrating the technical feasibility of the proposed approach. In addition, the project includes the development of an initial business plan, addressing market analysis, revenue model definition, user acquisition strategies, and financial projections, which indicate the potential economic viability of the solution. It is concluded that Omnivest fulfills the proposed objectives by validating the application of artificial intelligence and computer engineering techniques in the context of financial decision support and by establishing a solid foundation for future improvements and expansion of the platform.

Keywords : artificial intelligence; fundamental analysis; financial market; computer engineering; decision support systems.

List of Abbreviations and Acronyms

AI – Artificial Intelligence

API – Application Programming Interface

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1 Introduction

1.1 Context and Motivation

The growth of digital investment platforms and the increasing participation of individual investors in the Brazilian capital market have intensified the demand for accessible and reliable tools to support financial decision-making. Although a large volume of financial information is publicly available through corporate reports and market data providers, such information is often complex, fragmented, and difficult to interpret, especially for non-professional investors. In this context, technology plays a fundamental role in reducing informational barriers and enabling more informed investment decisions.

1.2 Problem Definition and Value Proposition

The central problem addressed by this work is the difficulty faced by retail investors in understanding financial statements, comparing companies across sectors and time periods, and evaluating the potential future performance of stocks. These challenges often lead to decisions based on incomplete information or subjective criteria. Omnivest proposes a value proposition centered on the integration of automated data extraction, standardized financial metrics, and predictive models into a single platform, allowing users to perform fundamental analysis and explore investment scenarios with greater efficiency and confidence.

1.3 Objectives of the Work

The general objective of this work is to design, implement, and validate a computational solution accompanied by a business plan for its introduction into the market. The specific objectives include the development of a full-stack web application, the construction of a data pipeline for processing corporate financial reports, the application of machine learning models for stock price forecasting, and the evaluation of the technical and economic feasibility of the proposed solution.

1.4 Justification and Contributions

This project is justified by its academic relevance, as it integrates concepts from computer engineering, data engineering, and artificial intelligence in a real-world application, and by its practical relevance, as it addresses a concrete market need within the financial sector. The main contributions of this work include the development of a modular and scalable system architecture, a structured approach to financial data processing, and an initial business model that connects technical innovation with market viability.

1.5 Work Structure

The remainder of this document is organized as follows. Chapter 2 presents the development of the proposed solution, including market analysis, technological implementation, business planning, and validation results. Chapter 3 presents the conclusions of the work and discusses future perspectives.

2 Solution Development

2.1 Definition of Market Assumptions and Hypotheses:

2.1.1 Problem Hypothesis

Retail investors in Brazil face challenges in understanding complex financial statements, comparing companies across time periods, and evaluating the future performance of stocks. These investors often rely on fragmented or superficial information, leading to suboptimal investment decisions. **Omnivest** assumes that these investors are willing to pay for a platform that simplifies and automates this process, providing them with actionable insights based on solid financial data.

2.1.2 Solution Hypothesis

Omnivest assumes that the best solution to address the pain points of retail investors is a data-driven, AI-powered platform that automates the extraction, standardization, and analysis of financial data. By integrating machine learning models to predict stock prices and offering clear, actionable visualizations, Omnivest aims to empower investors to make more informed decisions in a fraction of the time.

2.1.3 Value Hypothesis

Omnivest assumes that retail investors will find value in the platform's subscription model, as it delivers significant time savings and data-driven insights for their investment decisions. With a scalable pricing structure (monthly, biannual, and annual), Omnivest believes it can provide users with a clear return on investment (ROI) while meeting their specific needs in the financial market.

2.2 Market Sizing and Analysis:

2.2.1 Market Size (TAM, SAM, SOM):

- The Brazilian retail investment market has experienced significant growth over the last decade, driven by increased digitalization and broader access to the capital market. The Total Addressable Market (TAM) is defined as all individual investors who directly invest in equities through B3 (the Brazilian Stock Exchange). According to publicly available market data, this group is estimated at approximately 15 million individuals.
- The Serviceable Available Market (SAM) represents the subset of this population that actively seeks digital tools to support investment decision-making. This segment includes digitally literate investors who already use online brokerage platforms, financial applications, or analytical tools. Based on market reports and industry benchmarks, this group is estimated to range between 3 and 5 million potential users.

- The Serviceable Obtainable Market (SOM) corresponds to the portion of the SAM that Omnivest can realistically capture during its initial growth phase. Considering the early-stage nature of the platform, limited brand recognition, and an organic growth strategy, the SOM is conservatively estimated at approximately 50,000 to 150,000 users within the first 24 months of operation. This estimate focuses on early adopters and financially engaged users who actively seek data-driven investment solutions.

2.2.2 Customer Segmentation and Profiling

Based on qualitative research and market analysis, three primary personas were identified to guide the development and positioning of the Omnivest platform. These personas represent distinct user profiles with different motivations, pain points, and expectations, allowing the solution to address multiple segments within the Brazilian investment ecosystem.

The first persona, Rafael, represents the individual retail investor at an early stage of his investment journey. Rafael is typically a young professional with limited financial expertise who actively seeks accessible and trustworthy information to support his investment decisions. His main pain points include difficulty interpreting financial statements, uncertainty about investment choices, and lack of confidence in decision-making. Rafael values clarity, simplicity, and educational support, and is motivated by tools that help him reduce risk and make more informed choices without requiring deep technical knowledge. Omnivest addresses Rafael's needs by providing automated financial insights, intuitive dashboards, and predictive indicators that simplify complex data.

The second persona, Kauan, represents the financially engaged user with an intermediate level of investment knowledge. This profile includes individuals who already invest regularly and actively seek tools that allow deeper analysis and comparison between companies. Kauan values data accuracy, advanced metrics, and efficiency in analysis, as he often evaluates multiple assets simultaneously. His main motivation is optimizing investment decisions and improving portfolio performance. For this persona, Omnivest offers structured financial indicators,

comparative analysis tools, and AI-based forecasting models that support more sophisticated decision-making without the cost or complexity of professional-grade platforms.

The third persona, Eduardo, represents a business-oriented or executive profile, such as founders, managers, or decision-makers interested in financial performance analysis for strategic purposes. Eduardo is less focused on daily trading and more concerned with long-term value creation, company fundamentals, and macro-level insights. His pain points include fragmented data sources and lack of consolidated, reliable information to support strategic evaluations. Omnivest addresses this profile by offering standardized financial data, historical performance analysis, and predictive insights that support strategic planning and high-level financial assessment.

Together, these personas define Omnivest's target audience as digitally literate individuals who seek data-driven financial insights, ranging from beginner investors to experienced users and business decision-makers. This segmentation directly informs product design, pricing strategy, and communication tone, ensuring alignment between user expectations and platform capabilities.

2.3 Competitive Analysis and Differentials:

The competitive landscape in which Omnivest is positioned includes both high-end professional platforms and widely accessible informational tools, each addressing different segments of the financial analysis market. The analysis of direct and indirect competitors highlights gaps that motivate the proposed solution and clarify its competitive positioning.

One of the main reference competitors is the Bloomberg Terminal, a professional-grade financial information and analytics platform widely used by institutional investors and financial specialists. Bloomberg offers comprehensive, reliable, and real-time data, as well as advanced analytical tools. However, its high cost and operational complexity make it inaccessible to most retail investors,

particularly beginners. The steep learning curve and subscription pricing position Bloomberg outside the reach of Omnivest's primary target audience, creating an opportunity for more accessible alternatives.

Another relevant competitor is Investopedia, which stands out for its high traffic volume and extensive educational content related to finance and investments. While Investopedia is effective as a learning resource, it does not provide practical, actionable tools for financial decision-making. Its content-oriented approach lacks automated analysis, predictive capabilities, and personalized insights, limiting its usefulness for users seeking direct support in investment evaluation.

TradingView represents a platform with strong visual appeal and advanced charting capabilities, widely adopted by individual investors and traders. It offers technical analysis tools and community-driven insights but relies heavily on user interpretation. TradingView does not provide automated fundamental analysis or predictive models based on corporate financial data, which restricts its ability to support long-term, data-driven investment decisions for less experienced users.

In contrast, Omnivest positions itself as a solution that bridges the gap between professional analytics platforms and purely informational or visualization-based tools. Its primary competitive differentiator lies in the integration of automated financial data extraction, standardized fundamental metrics, and artificial intelligence-based predictive models within a single, user-friendly platform. Unlike Bloomberg, Omnivest focuses on accessibility and affordability; unlike Investopedia, it delivers actionable insights rather than educational content alone; and unlike TradingView, it emphasizes automation and prediction rather than manual interpretation.

Therefore, Omnivest's competitive advantage is defined by its ability to democratize advanced financial analysis, offering predictive intelligence and structured insights tailored to retail investors and business-oriented users who seek clarity, efficiency, and confidence in their decision-making processes.

2.4 Technological Solution

2.4.1 Requirements and Specifications:

- Functional Requirements
 - User registration and authentication (OAuth2, JWT tokens)
 - Data visualization for financial metrics (charts, tables, predictions)
 - Company data extraction from official reports (PDF to JSON conversion)
 - Predictive analytics for stock price forecasting
- Non-Functional Requirements
 - Scalability to handle growing data and user base
 - High availability to support continuous market analysis
 - Data security for user and financial data protection
- User Specifications and Use Cases
 - Retail investor logs into the platform, selects companies, and views financial metrics.
 - Retail investor requests stock price predictions for the next quarter, based on historical data and financial reports.

2.4.2 Architecture and Technology:

- Omnivest follows a client–server architecture with modular backend components.
 - Frontend: React with TailwindCSS for responsive design
 - Backend: FastAPI for fast, efficient APIs
 - Data storage: PostgreSQL (financial data)
 - Predictive models: PyCaret (machine learning) for regression models

2.4.3 Development and Implementation (MVP):

- Methodology: Agile development with **Scrum** methodology, delivering functional iterations every 2 weeks.
- MVP Features
 - User authentication
 - Data retrieval and display for 3–5 selected companies
 - Stock price prediction model using historical data

2.4.4 Testing and Technical Evaluation:

- Unit Testing: All critical backend services are unit-tested (FastAPI).
- Integration Testing: End-to-end testing of the data pipeline and prediction model.
- Acceptance Testing: User testing with a small group of target customers to refine UX/UI and validate the predictive features.

2.5 The Business Plan

2.5.1 Market and Competitor Analysis:

The market targeted by Omnivest consists primarily of individual investors operating in the Brazilian capital market who seek accessible, reliable, and data-driven tools to support financial decision-making. Based on qualitative research, three main customer profiles were identified: beginner retail investors, intermediate investors with regular market activity, and business-oriented users interested in long-term financial performance analysis. These segments differ in experience and objectives but share a common need for clarity, efficiency, and structured financial information.

The competitive environment includes both professional-grade platforms and widely accessible informational tools. High-end solutions such as the Bloomberg Terminal provide comprehensive and reliable financial data but are characterized by high costs and operational complexity, which restrict their use to institutional investors. On the other hand, platforms such as Investopedia offer extensive educational content but lack actionable tools and automated analysis. TradingView provides advanced visualizations and charting features; however, it relies heavily on manual interpretation and does not offer automated fundamental analysis or predictive models. This landscape reveals a gap between professional analytics platforms and tools designed for non-specialist users.

A SWOT analysis was conducted to assess Omnivest's strategic positioning. The platform's main strengths include the integration of artificial intelligence for automated financial analysis, a focus on accessibility for non-professional investors, and a modular technical architecture. Identified weaknesses include limited brand recognition and dependence on the quality of public financial data. Opportunities arise from the rapid growth of retail investors in Brazil and the increasing demand for affordable analytical tools. Potential threats include competition from established financial platforms and changes in regulatory requirements within the financial sector.

2.5.2 Business Model (Business Model Canvas - BMC):

The business model of Omnivest was structured using the Business Model Canvas (BMC) framework. The value proposition centers on delivering automated, AI-driven financial insights that transform complex public data into structured and actionable information. Customer segments include beginner and intermediate retail investors, as well as business-oriented users seeking long-term performance indicators.

Customer relationships are maintained through a subscription-based model that provides continuous access to dashboards, predictive analytics, and updated financial indicators. Revenue streams are generated through recurring subscription plans with different billing periods, allowing users to choose the option that best fits their usage profile. Key activities involve data extraction from public sources, data processing and standardization, training and maintenance of predictive models, and platform development. Key resources include the software platform, machine learning models, and structured financial datasets. The cost structure is primarily composed of development efforts, infrastructure usage, and marketing activities.

2.5.3 Marketing and Sales Strategy:

The marketing and sales strategy of Omnivest follows a digital-first approach, focusing on reaching potential users through online channels aligned with their information-seeking behavior. The go-to-market strategy emphasizes transparency and clarity rather than traditional financial education, positioning the platform as an intelligent assistant that simplifies investment analysis.

Customer acquisition strategies include performance-based digital advertising, content distribution through financial and technology-oriented channels, partnerships with influencers in the personal finance domain, and the use of a public financial data dashboard as a lead-generation mechanism. This dashboard allows potential users to interact with structured financial information before subscribing, reducing adoption barriers. Customer retention strategies focus on delivering continuous value through

personalized insights, regular updates to predictive models, and a subscription structure designed to encourage long-term engagement.

2.5.4 Financial Projection and Feasibility:

The financial analysis conducted for Omnivest aims to assess the economic feasibility of the proposed solution rather than to seek external investment. The platform adopts a subscription-based revenue model with pricing designed to be accessible to the target audience while supporting sustainable operation. Projected expenses include infrastructure usage, development activities, and marketing efforts.

Based on conservative adoption assumptions, the analysis indicates that the platform has the potential to reach a break-even point within approximately 24 months of operation. Viability indicators such as recurring revenue growth and customer retention rates suggest that the business can scale gradually through reinvestment of generated revenue. No external investment is required at this stage, as the project is designed to grow organically while validating its value proposition in the market.

2.6 Validation and Results

2.6.1 Validation Methodology:

The validation approach adopted in this project was exploratory and qualitative in nature, which is consistent with the early development stage of the proposed solution and its primary focus on technical feasibility. Rather than conducting formal interviews or large-scale surveys, informal exploratory conversations were carried out with potential users to better understand their needs, expectations, and perceived pain points related to financial analysis tools. These interactions aimed to validate core assumptions regarding usability, perceived value, and feature relevance, rather

than to generate statistically representative results. In parallel, the functional prototype was evaluated through internal testing and limited user interaction to verify the coherence between the identified user needs and the implemented features. This validation strategy allowed the project to confirm the alignment between the proposed value proposition and user expectations while acknowledging that broader quantitative validation is a subject for future work.

2.6.2 Market Validation Results:

Feedback obtained from potential users during exploratory interactions indicated a strong interest in features related to automated financial analysis and stock price prediction. Participants highlighted the importance of clarity, ease of use, and the ability to compare companies efficiently.

Additionally, preliminary engagement metrics from prototype usage suggest positive user interest, with a subset of users indicating willingness to continue using the platform as additional features are developed.

2.6.3 Key Performance Indicators (KPIs):

The following indicators are preliminary estimates based on early validation assumptions and market benchmarks, intended to support feasibility analysis rather than to represent observed operational metrics.

- Customer Acquisition Cost (CAC): Estimated at R\$50 per user during the initial growth phase, considering performance-based digital marketing strategies.
- Lifetime Value (LTV): Estimated at approximately R\$500 per user over a one-year period, based on the projected subscription pricing model and expected user retention.
- Churn Rate: Estimated at 10% after six months, reflecting conservative assumptions for early-stage digital platforms in the financial technology sector.

2.6.4 Risks and Mitigation Plan:

- Regulatory changes in the financial sector (Mitigation: Consult with legal experts)
- Low user retention (Mitigation: Improve onboarding experience, offer personalized reports)

3 Conclusion

This project successfully designed and implemented Omnivest, a platform that addresses the need for accessible financial analysis tools in the Brazilian market. The objectives were achieved through the development of a full-stack web application, the application of AI-based forecasting models, and the creation of a business plan that demonstrates market viability. Future work will focus on expanding the platform's features and optimizing the prediction models.

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